This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-CT-7801-01** Sheet 1 of 1

OPL Title	**Barscreen Cleaning (Suction Screen)**	**Discipline**	Process / Operations
Equipment	CT-7801 - COOLING TOWER CELL FAN	**Area / Unit**	7800 - COOLING TOWER SYSTEM
Related Interlock	SEQ-7801 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-7801

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **barscreen cleaning (suction screen)** for CT-7801 (COOLING TOWER CELL FAN) in the COOLING TOWER SYSTEM. A clogged barscreen starves the circulating water pumps and fouls the fill, reducing cooling for the whole unit. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **NON CRITICAL** handling **Circulating water / air** at up to atm / 50 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID;
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Rake the SS304 barscreen every shift on the round	Permit valid & briefed
2	Remove leaves, debris and biological growth	Parameter within limit
3	Check the differential level across the screen	Interlock status confirmed
4	Dispose of debris correctly and log the cleaning	Change logged on DCS
5	Report rapid re-fouling to water treatment	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Routine barscreen cleaning on operator round	Normal debris accumulation	Raked barscreen clear, checked differential level
Scheduled fan gearbox oil change	Routine - oil sample acceptable	Drained and refilled ISO VG 220, replaced breather
VSHH-7802 vibration alarm on CT-7801 fan	Leading-edge erosion on one FRP blade caused imbalance	Inspected blades, re-balanced fan, vibration normalised

6. KEY LEARNING POINTS

- A clogged barscreen starves the circulating water pumps and fouls the fill, reducing cooling for the whole unit.
- Respect the interlock SEQ-7801: e.g. VSHH-7802 trips at > 9 mm/s - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-7801 before working on CT-7801.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Arya Wibisono (EMP-0912)	Wednesday, 27 May 2026

Cross-ref tag CT-7801 across Datasheet, GA, P&ID, Interlock and Maintenance History

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-CT-7801-02** Sheet 1 of 1

OPL Title	**Gearbox Oil Level & Change (ISO VG 220)**	**Discipline**	Mechanical
Equipment	CT-7801 - COOLING TOWER CELL FAN	**Area / Unit**	7800 - COOLING TOWER SYSTEM
Related Interlock	SEQ-7801 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-7801

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **gearbox oil level & change (iso vg 220)** for CT-7801 (COOLING TOWER CELL FAN) in the COOLING TOWER SYSTEM. Low or degraded gearbox oil causes overheating (TSHH-7803) and gear failure high up on the fan deck. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **NON CRITICAL** handling **Circulating water / air** at up to atm / 50 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Check the gearbox sight-glass oil level	Isolation & LOTO verified
2	Sample the oil for water contamination and metal	Parts match BOM / spec
3	Change the ISO VG 220 (4.5 L) per the schedule	Torque / clearance to spec
4	Verify the oil pressure PSL-7807 after refill	No leak / smooth operation
5	Record the oil change and next due date	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Scheduled fan gearbox oil change	Routine - oil sample acceptable	Drained and refilled ISO VG 220, replaced breather
VSHH-7802 vibration alarm on CT-7801 fan	Leading-edge erosion on one FRP blade caused imbalance	Inspected blades, re-balanced fan, vibration normalised
TSHH-7803 gearbox oil temperature rising	Degraded oil and fouled oil cooler	Changed oil, cleaned oil cooler, verified temperature

6. KEY LEARNING POINTS

- Low or degraded gearbox oil causes overheating (TSHH-7803) and gear failure high up on the fan deck.
- Respect the interlock SEQ-7801: e.g. VSHH-7802 trips at > 9 mm/s - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-7801 before working on CT-7801.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Arya Wibisono (EMP-0912)	Sunday, 31 May 2026

Cross-ref tag CT-7801 across Datasheet, GA, P&ID, Interlock and Maintenance History

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-CT-7801-03** Sheet 1 of 1

OPL Title	**Fan Blade Inspection & Balance (FRP)**	**Discipline**	Mechanical
Equipment	CT-7801 - COOLING TOWER CELL FAN	**Area / Unit**	7800 - COOLING TOWER SYSTEM
Related Interlock	SEQ-7801 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-7801

Classification:	[ ] Basic Knowledge	[X] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **fan blade inspection & balance (frp)** for CT-7801 (COOLING TOWER CELL FAN) in the COOLING TOWER SYSTEM. A cracked or eroded FRP blade unbalances the fan and causes the VSHH-7802 vibration trip. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **NON CRITICAL** handling **Circulating water / air** at up to atm / 50 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Lock out the fan and inspect all 6 FRP blades	Isolation & LOTO verified
2	Check for cracks, leading-edge erosion and tip clearance	Parts match BOM / spec
3	Verify the blade pitch angle is uniform on all blades	Torque / clearance to spec
4	Re-balance the fan if vibration is high	No leak / smooth operation
5	Torque the blade root bolts to specification	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Scheduled fan gearbox oil change	Routine - oil sample acceptable	Drained and refilled ISO VG 220, replaced breather
VSHH-7802 vibration alarm on CT-7801 fan	Leading-edge erosion on one FRP blade caused imbalance	Inspected blades, re-balanced fan, vibration normalised
TSHH-7803 gearbox oil temperature rising	Degraded oil and fouled oil cooler	Changed oil, cleaned oil cooler, verified temperature

6. KEY LEARNING POINTS

- A cracked or eroded FRP blade unbalances the fan and causes the VSHH-7802 vibration trip.
- Respect the interlock SEQ-7801: e.g. VSHH-7802 trips at > 9 mm/s - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-7801 before working on CT-7801.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Arya Wibisono (EMP-0912)	Thursday, 04 June 2026

Cross-ref tag CT-7801 across Datasheet, GA, P&ID, Interlock and Maintenance History

This is sample data provided for CALIBER purposes only	ONE POINT LESSON (OPL)	OPL No: **OPL-CT-7801-04** Sheet 1 of 1

OPL Title	**Circulating Water pH Control (AI-7805)**	**Discipline**	Process / Operations
Equipment	CT-7801 - COOLING TOWER CELL FAN	**Area / Unit**	7800 - COOLING TOWER SYSTEM
Related Interlock	SEQ-7801 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-7801

Classification:	[ ] Basic Knowledge	[X] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **circulating water ph control (ai-7805)** for CT-7801 (COOLING TOWER CELL FAN) in the COOLING TOWER SYSTEM. Wrong pH causes scaling or corrosion across the entire cooling water loop, not just this cell. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **NON CRITICAL** handling **Circulating water / air** at up to atm / 50 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID;
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Read the AI-7805 pH each shift (target 7.0-8.5)	Permit valid & briefed
2	Check the acid / caustic dosing pumps are running	Parameter within limit
3	Cross-check against a manual lab titration	Interlock status confirmed
4	Adjust the dosing set-point as needed	Change logged on DCS
5	Report large pH swings to water treatment	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Routine barscreen cleaning on operator round	Normal debris accumulation	Raked barscreen clear, checked differential level
Scheduled fan gearbox oil change	Routine - oil sample acceptable	Drained and refilled ISO VG 220, replaced breather
VSHH-7802 vibration alarm on CT-7801 fan	Leading-edge erosion on one FRP blade caused imbalance	Inspected blades, re-balanced fan, vibration normalised

6. KEY LEARNING POINTS

- Wrong pH causes scaling or corrosion across the entire cooling water loop, not just this cell.
- Respect the interlock SEQ-7801: e.g. VSHH-7802 trips at > 9 mm/s - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-7801 before working on CT-7801.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Arya Wibisono (EMP-0912)	Monday, 08 June 2026

Cross-ref tag CT-7801 across Datasheet, GA, P&ID, Interlock and Maintenance History

This is sample data provided for CALIBER purposes only	ONE POINT LESSON (OPL)	OPL No: **OPL-CT-7801-05** Sheet 1 of 1

OPL Title	**Drift Eliminator & Fill Inspection**	**Discipline**	Mechanical
Equipment	CT-7801 - COOLING TOWER CELL FAN	**Area / Unit**	7800 - COOLING TOWER SYSTEM
Related Interlock	SEQ-7801 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-7801

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **drift eliminator & fill inspection** for CT-7801 (COOLING TOWER CELL FAN) in the COOLING TOWER SYSTEM. Fouled or collapsed fill and drift eliminators cut the cooling performance and raise the basin temperature. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **NON CRITICAL** handling **Circulating water / air** at up to atm / 50 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Inspect the PVC fill for scaling and collapse	Isolation & LOTO verified
2	Check the drift eliminator packs are aligned	Parts match BOM / spec
3	Hose down fouled sections during a cell outage	Torque / clearance to spec
4	Report a thermal performance drop via TT-7801	No leak / smooth operation
5	Plan fill replacement if degradation is widespread	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Scheduled fan gearbox oil change	Routine - oil sample acceptable	Drained and refilled ISO VG 220, replaced breather
VSHH-7802 vibration alarm on CT-7801 fan	Leading-edge erosion on one FRP blade caused imbalance	Inspected blades, re-balanced fan, vibration normalised
TSHH-7803 gearbox oil temperature rising	Degraded oil and fouled oil cooler	Changed oil, cleaned oil cooler, verified temperature

6. KEY LEARNING POINTS

- Fouled or collapsed fill and drift eliminators cut the cooling performance and raise the basin temperature.
- Respect the interlock SEQ-7801: e.g. VSHH-7802 trips at > 9 mm/s - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-7801 before working on CT-7801.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Arya Wibisono (EMP-0912)	Friday, 12 June 2026

Cross-ref tag CT-7801 across Datasheet, GA, P&ID, Interlock and Maintenance History

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-CT-7801-06** Sheet 1 of 1

OPL Title	**Gearbox Vibration Monitoring (VSHH-7802)**	**Discipline**	Instrument
Equipment	CT-7801 - COOLING TOWER CELL FAN	**Area / Unit**	7800 - COOLING TOWER SYSTEM
Related Interlock	SEQ-7801 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-7801

Classification:	[ ] Basic Knowledge	[X] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **gearbox vibration monitoring (vshh-7802)** for CT-7801 (COOLING TOWER CELL FAN) in the COOLING TOWER SYSTEM. A rising vibration trend warns of gear or bearing wear before the fan trips or throws a blade. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **NON CRITICAL** handling **Circulating water / air** at up to atm / 50 degC. Observe the following before and during the task:

- De-energise and isolate the loop; place the function to bypass/inhibit under permit.
- Inform the panel operator before working on any trip, valve or transmitter.
- Take care with instrument air / hydraulic pressure when disconnecting tubing.
- Verify intrinsic-safety / Ex integrity is restored before returning to service.
- Wear correct PPE and follow ESD/override control procedures.

3. TOOLS & MATERIALS REQUIRED

- HART communicator / positioner software (e.g. ValveLink)
- Loop calibrator & decade box / pressure source
- Digital multimeter and clamp meter
- Span/zero test gas or reference standards
- Torque screwdriver and small hand tools
- Certified test/inhibit permit

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Trend the VS-7802 reading on the PdM route	Loop isolated / under permit
2	Alarm at 6 mm/s and trip at 9 mm/s (VSHH-7802)	Reference standard traceable
3	Correlate the rise with gearbox oil temperature TE-7803	As-found & as-left recorded
4	Inspect the floating drive-shaft couplings	Trip / control action verified
5	Escalate a step-change to reliability	Return to service confirmed

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Scheduled pH analyzer calibration	Electrode drift	Recalibrated AI-7805 on buffer solutions, replaced electrode
Blowdown conductivity reading unreliable	Conductivity cell fouled with scale	Cleaned and recalibrated AI-7806 conductivity cell
Routine barscreen cleaning on operator round	Normal debris accumulation	Raked barscreen clear, checked differential level

6. KEY LEARNING POINTS

- A rising vibration trend warns of gear or bearing wear before the fan trips or throws a blade.
- Respect the interlock SEQ-7801: e.g. VSHH-7802 trips at > 9 mm/s - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-7801 before working on CT-7801.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Arya Wibisono (EMP-0912)	Tuesday, 16 June 2026

Cross-ref tag CT-7801 across Datasheet, GA, P&ID, Interlock and Maintenance History

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-CT-7801-07** Sheet 1 of 1

OPL Title	**Conductivity / Blowdown Control (AI-7806)**	**Discipline**	Process / Operations
Equipment	CT-7801 - COOLING TOWER CELL FAN	**Area / Unit**	7800 - COOLING TOWER SYSTEM
Related Interlock	SEQ-7801 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-7801

Classification:	[ ] Basic Knowledge	[X] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **conductivity / blowdown control (ai-7806)** for CT-7801 (COOLING TOWER CELL FAN) in the COOLING TOWER SYSTEM. High conductivity means the water is over-concentrated; automatic blowdown keeps the cycles of concentration in range. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **NON CRITICAL** handling **Circulating water / air** at up to atm / 50 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID;
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Read the AI-7806 conductivity value	Permit valid & briefed
2	Verify the automatic blowdown valve XV-7806 operates	Parameter within limit
3	Maintain the target cycles of concentration	Interlock status confirmed
4	Log make-up versus blowdown volumes	Change logged on DCS
5	Coordinate chemical dosing with water treatment	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Routine barscreen cleaning on operator round	Normal debris accumulation	Raked barscreen clear, checked differential level
Scheduled fan gearbox oil change	Routine - oil sample acceptable	Drained and refilled ISO VG 220, replaced breather
VSHH-7802 vibration alarm on CT-7801 fan	Leading-edge erosion on one FRP blade caused imbalance	Inspected blades, re-balanced fan, vibration normalised

6. KEY LEARNING POINTS

- High conductivity means the water is over-concentrated; automatic blowdown keeps the cycles of concentration in range.
- Respect the interlock SEQ-7801: e.g. VSHH-7802 trips at > 9 mm/s - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-7801 before working on CT-7801.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Arya Wibisono (EMP-0912)	Saturday, 20 June 2026

Cross-ref tag CT-7801 across Datasheet, GA, P&ID, Interlock and Maintenance History

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-FA-8901-01** Sheet 1 of 1

OPL Title	**Level Gauge / Bridle Verification (LT-8901)**	**Discipline**	Instrument
Equipment	FA-8901 - REFLUX ACCUMULATOR DRUM	**Area / Unit**	8900 - HEXANE COLUMN OVERHEAD SYSTEM
Related Interlock	SEQ-8901 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-8901

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **level gauge / bridle verification (lt-8901)** for FA-8901 (REFLUX ACCUMULATOR DRUM) in the HEXANE COLUMN OVERHEAD SYSTEM. A false level reading causes reflux pump cavitation (low) or column flooding (high); the transmitter must agree with the gauge. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Hexane / water** at up to 10 barg / FV / 120 degC. Observe the following before and during the task:

- De-energise and isolate the loop; place the function to bypass/inhibit under permit.
- Inform the panel operator before working on any trip, valve or transmitter.
- Take care with instrument air / hydraulic pressure when disconnecting tubing.
- Verify intrinsic-safety / Ex integrity is restored before returning to service.
- Wear correct PPE and follow ESD/override control procedures.

3. TOOLS & MATERIALS REQUIRED

- HART communicator / positioner software (e.g. ValveLink)
- Loop calibrator & decade box / pressure source
- Digital multimeter and clamp meter
- Span/zero test gas or reference standards
- Torque screwdriver and small hand tools
- Certified test/inhibit permit

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Cross-check LT-8901 against the level bridle gauge	Loop isolated / under permit
2	Blow down the bridle to clear any plugging	Reference standard traceable
3	Confirm the LSHH-8901 and LSLL-8901 switch actions	As-found & as-left recorded
4	Verify the level tracks correctly through a level change	Trip / control action verified
5	Log any deviation and recalibrate if needed	Return to service confirmed

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Reflux drum level transmitter drifting vs gauge	Transmitter zero drift	Recalibrated LT-8901 against the bridle gauge
Blanket / vent split-range valve hunting	Split-range calibration overlap incorrect	Re-tuned PCV-8905 split-range, verified PT-8902 hold
Proof test of drum high-high level trip	Within tolerance	Recalibrated LSHH-8901 and verified EFF-1 column feed trip

6. KEY LEARNING POINTS

- A false level reading causes reflux pump cavitation (low) or column flooding (high); the transmitter must agree with the gauge.
- Respect the interlock SEQ-8901: e.g. LSHH-8901 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-8901 before working on FA-8901.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Zulkarnain Hasan (EMP-0901)	Sunday, 07 June 2026

Cross-ref tag FA-8901 across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	ONE POINT LESSON (OPL)	OPL No: **OPL-FA-8901-02** Sheet 1 of 1

OPL Title	**PSV-8901 Testing & Car-Seal Management**	**Discipline**	Mechanical
Equipment	FA-8901 - REFLUX ACCUMULATOR DRUM	**Area / Unit**	8900 - HEXANE COLUMN OVERHEAD SYSTEM
Related Interlock	SEQ-8901 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-8901

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **psv-8901 testing & car-seal management** for FA-8901 (REFLUX ACCUMULATOR DRUM) in the HEXANE COLUMN OVERHEAD SYSTEM. The relief valve is the drum's last line of overpressure defence and must be proven and never isolated in service. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Hexane / water** at up to 10 barg / FV / 120 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Confirm PSV-8901 is set to 10 barg	Isolation & LOTO verified
2	Carry out the off-line pop test per the statutory schedule	Parts match BOM / spec
3	Keep the PSV isolation car-sealed open in service	Torque / clearance to spec
4	Tag and file the test certificate	No leak / smooth operation
5	Never block-in a live PSV without an authorised spare	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Statutory relief valve test	Set pressure drift found	Overhauled PSV-8901, lapped seat, re-set 10 barg, certified
Boot drain valve N4 passing / leaking	Seat eroded by water/solids	Reseated / replaced boot drain valve HV-8904
Scheduled corrosion monitoring of boot	Wall thickness within corrosion allowance	UT survey of boot and shell bottom, logged in RBI file

6. KEY LEARNING POINTS

- The relief valve is the drum's last line of overpressure defence and must be proven and never isolated in service.
- Respect the interlock SEQ-8901: e.g. LSHH-8901 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-8901 before working on FA-8901.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Zulkarnain Hasan (EMP-0901)	Thursday, 11 June 2026

Cross-ref tag FA-8901 across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-FA-8901-03** Sheet 1 of 1

OPL Title	**Water Boot Draining (LT-8903)**	**Discipline**	Process / Operations
Equipment	FA-8901 - REFLUX ACCUMULATOR DRUM	**Area / Unit**	8900 - HEXANE COLUMN OVERHEAD SYSTEM
Related Interlock	SEQ-8901 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-8901

Classification:	[ ] Basic Knowledge	[ ] Improvement	[X] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **water boot draining (lt-8903)** for FA-8901 (REFLUX ACCUMULATOR DRUM) in the HEXANE COLUMN OVERHEAD SYSTEM. Water accumulating in the boot upsets reflux quality and can slug the reflux pump; regular draining keeps the interface controlled. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Hexane / water** at up to 10 barg / FV / 120 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID;
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Monitor the boot interface level LT-8903	Permit valid & briefed
2	Drain the boot N4 to the oily-water sewer via HV-8904	Parameter within limit
3	Stop draining before hydrocarbon breaks through (watch interface)	Interlock status confirmed
4	Log the drain volume each shift	Change logged on DCS
5	Report a rising water make to process	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Reflux drum level transmitter drifting vs gauge	Transmitter zero drift	Recalibrated LT-8901 against the bridle gauge
Statutory relief valve test	Set pressure drift found	Overhauled PSV-8901, lapped seat, re-set 10 barg, certified
Boot drain valve N4 passing / leaking	Seat eroded by water/solids	Reseated / replaced boot drain valve HV-8904

6. KEY LEARNING POINTS

- Water accumulating in the boot upsets reflux quality and can slug the reflux pump; regular draining keeps the interface controlled.
- Respect the interlock SEQ-8901: e.g. LSHH-8901 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-8901 before working on FA-8901.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Zulkarnain Hasan (EMP-0901)	Monday, 15 June 2026

Cross-ref tag FA-8901 across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-FA-8901-04** Sheet 1 of 1

OPL Title	**Nitrogen Blanketing (PCV-8905)**	**Discipline**	Process / Operations
Equipment	FA-8901 - REFLUX ACCUMULATOR DRUM	**Area / Unit**	8900 - HEXANE COLUMN OVERHEAD SYSTEM
Related Interlock	SEQ-8901 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-8901

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **nitrogen blanketing (pcv-8905)** for FA-8901 (REFLUX ACCUMULATOR DRUM) in the HEXANE COLUMN OVERHEAD SYSTEM. The nitrogen blanket keeps air out of the hexane drum and holds a stable pressure; loss of blanket risks an explosive atmosphere. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Hexane / water** at up to 10 barg / FV / 120 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID;
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Verify the PCV-8905 split-range holds PT-8902 pressure	Permit valid & briefed
2	Confirm the blanket makes up on falling pressure	Parameter within limit
3	Confirm the vent relieves on rising pressure	Interlock status confirmed
4	Respond immediately to a blanket-loss alarm	Change logged on DCS
5	Check the N2 supply header pressure is adequate	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Reflux drum level transmitter drifting vs gauge	Transmitter zero drift	Recalibrated LT-8901 against the bridle gauge
Statutory relief valve test	Set pressure drift found	Overhauled PSV-8901, lapped seat, re-set 10 barg, certified
Boot drain valve N4 passing / leaking	Seat eroded by water/solids	Reseated / replaced boot drain valve HV-8904

6. KEY LEARNING POINTS

- The nitrogen blanket keeps air out of the hexane drum and holds a stable pressure; loss of blanket risks an explosive atmosphere.
- Respect the interlock SEQ-8901: e.g. LSHH-8901 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-8901 before working on FA-8901.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Zulkarnain Hasan (EMP-0901)	Friday, 19 June 2026

Cross-ref tag FA-8901 across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	ONE POINT LESSON (OPL)	OPL No: **OPL-FA-8901-05** Sheet 1 of 1

OPL Title	**Reflux Pump Low-Level Protection (LSLL-8901)**	**Discipline**	Instrument
Equipment	FA-8901 - REFLUX ACCUMULATOR DRUM	**Area / Unit**	8900 - HEXANE COLUMN OVERHEAD SYSTEM
Related Interlock	SEQ-8901 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-8901

Classification:	[ ] Basic Knowledge	[ ] Improvement	[X] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **reflux pump low-level protection (lsll-8901)** for FA-8901 (REFLUX ACCUMULATOR DRUM) in the HEXANE COLUMN OVERHEAD SYSTEM. A low drum level cavitates and damages the GA-8920 reflux pump; the LSLL trip must be proven and never casually bypassed. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Hexane / water** at up to 10 barg / FV / 120 degC. Observe the following before and during the task:

- De-energise and isolate the loop; place the function to bypass/inhibit under permit.
- Inform the panel operator before working on any trip, valve or transmitter.
- Take care with instrument air / hydraulic pressure when disconnecting tubing.
- Verify intrinsic-safety / Ex integrity is restored before returning to service.
- Wear correct PPE and follow ESD/override control procedures.

3. TOOLS & MATERIALS REQUIRED

- HART communicator / positioner software (e.g. ValveLink)
- Loop calibrator & decade box / pressure source
- Digital multimeter and clamp meter
- Span/zero test gas or reference standards
- Torque screwdriver and small hand tools
- Certified test/inhibit permit

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Confirm the LSLL-8901 trip stops GA-8920	Loop isolated / under permit
2	Test the trip during commissioning and proof tests	Reference standard traceable
3	Never bypass the trip without an authorised permit	As-found & as-left recorded
4	Investigate any repeated low-level events	Trip / control action verified
5	Verify the pump restarts only after level recovers	Return to service confirmed

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Reflux drum level transmitter drifting vs gauge	Transmitter zero drift	Recalibrated LT-8901 against the bridle gauge
Blanket / vent split-range valve hunting	Split-range calibration overlap incorrect	Re-tuned PCV-8905 split-range, verified PT-8902 hold
Proof test of drum high-high level trip	Within tolerance	Recalibrated LSHH-8901 and verified EFF-1 column feed trip

6. KEY LEARNING POINTS

- A low drum level cavitates and damages the GA-8920 reflux pump; the LSLL trip must be proven and never casually bypassed.
- Respect the interlock SEQ-8901: e.g. LSHH-8901 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-8901 before working on FA-8901.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Zulkarnain Hasan (EMP-0901)	Tuesday, 23 June 2026

Cross-ref tag FA-8901 across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	ONE POINT LESSON (OPL)	OPL No: **OPL-FA-8901-06** Sheet 1 of 1

OPL Title	**Corrosion / Thickness Inspection**	**Discipline**	Mechanical
Equipment	FA-8901 - REFLUX ACCUMULATOR DRUM	**Area / Unit**	8900 - HEXANE COLUMN OVERHEAD SYSTEM
Related Interlock	SEQ-8901 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-8901

Classification:	[ ] Basic Knowledge	[X] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **corrosion / thickness inspection** for FA-8901 (REFLUX ACCUMULATOR DRUM) in the HEXANE COLUMN OVERHEAD SYSTEM. The boot and shell bottom see the water phase and are corrosion-prone; periodic thickness checks catch wall loss early. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Hexane / water** at up to 10 barg / FV / 120 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Take UT thickness readings on the shell bottom and boot	Isolation & LOTO verified
2	Compare readings to the 3.0 mm corrosion allowance	Parts match BOM / spec
3	Inspect the mist pad and vortex breaker at turnaround	Torque / clearance to spec
4	Record the readings in the RBI / inspection file	No leak / smooth operation
5	Flag any accelerated wall loss to inspection engineering	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Statutory relief valve test	Set pressure drift found	Overhauled PSV-8901, lapped seat, re-set 10 barg, certified
Boot drain valve N4 passing / leaking	Seat eroded by water/solids	Reseated / replaced boot drain valve HV-8904
Scheduled corrosion monitoring of boot	Wall thickness within corrosion allowance	UT survey of boot and shell bottom, logged in RBI file

6. KEY LEARNING POINTS

- The boot and shell bottom see the water phase and are corrosion-prone; periodic thickness checks catch wall loss early.
- Respect the interlock SEQ-8901: e.g. LSHH-8901 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-8901 before working on FA-8901.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Zulkarnain Hasan (EMP-0901)	Saturday, 27 June 2026

Cross-ref tag FA-8901 across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-FA-8901-07** Sheet 1 of 1

OPL Title	**Manway Box-Up After Internal Inspection**	**Discipline**	Mechanical
Equipment	FA-8901 - REFLUX ACCUMULATOR DRUM	**Area / Unit**	8900 - HEXANE COLUMN OVERHEAD SYSTEM
Related Interlock	SEQ-8901 (SIL 1)	**P&ID; Ref**	TJC-LLD-PID-8901

Classification:	[ ] Basic Knowledge	[X] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **manway box-up after internal inspection** for FA-8901 (REFLUX ACCUMULATOR DRUM) in the HEXANE COLUMN OVERHEAD SYSTEM. Correct gasket selection and bolt torque after inspection prevent a hexane leak from the manway. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Hexane / water** at up to 10 barg / FV / 120 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Fit a new SW SS316/graphite manway gasket	Isolation & LOTO verified
2	Torque the manway bolts in a star pattern in passes	Parts match BOM / spec
3	Leak test with the nitrogen blanket	Torque / clearance to spec
4	Re-instate the PSV and all instruments	No leak / smooth operation
5	Confirm the drum is ready before re-introducing hydrocarbon	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Statutory relief valve test	Set pressure drift found	Overhauled PSV-8901, lapped seat, re-set 10 barg, certified
Boot drain valve N4 passing / leaking	Seat eroded by water/solids	Reseated / replaced boot drain valve HV-8904
Scheduled corrosion monitoring of boot	Wall thickness within corrosion allowance	UT survey of boot and shell bottom, logged in RBI file

6. KEY LEARNING POINTS

- Correct gasket selection and bolt torque after inspection prevent a hexane leak from the manway.
- Respect the interlock SEQ-8901: e.g. LSHH-8901 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-8901 before working on FA-8901.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Zulkarnain Hasan (EMP-0901)	Wednesday, 01 July 2026

Cross-ref tag FA-8901 across Datasheet, GA, P&ID, Interlock and Maintenance History.